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import streamlit as st import math import heapq import networkx as nx import matplotlib.pyplot as plt # Graph, Use Case: Emergency Supply Robot locations = { "Pharmacy": (0, 0), "Main_Corridor": (2, 1), "Patient_Wing": (1, 4), "Nursing_Station": (4, 2), "Laboratory": (5, 5), "Emergency_Ward": (8, 6) } hospital_graph = { "Pharmacy": { "Main_Corridor": 2.2, "Patient_Wing": 4.1 }, "Main_Corridor": { "Nursing_Station": 2.2 }, "Patient_Wing": { "Laboratory": 5.0 }, "Nursing_Station": { "Laboratory": 3.2, "Emergency_Ward": 6.0 }, "Laboratory": { "Emergency_Ward": 3.2 }, "Emergency_Ward": {} } # Heuristic def heuristic(current, goal): pass # Path reconstruction def reconstruct_path(came_from, current): pass # GBFS def gbfs(start, goal): pass # A* def a_star(start, goal): pass
############################################## # Streamlit GUI Code # Set Page Config # write meaningful title and description for the app # define the nodes and their coordinates nodes = list(hospital_graph.keys()) # create a selectbox for the user to choose the start and goal nodes start = st.selectbox( "Select Initial Node", #pass the list of nodes to the selectbox # set the default value to "Pharmacy" ) goal = st.selectbox( "Select Goal Node", #pass the list of nodes to the selectbox # set the default value to "Emergency_Ward" ) # create a selectbox for the user to choose the search algorithm if st.button("Run Search"): if algorithm == "GBFS": # run the GBFS algorithm with the selected start and goal nodes pass else: # run the A* algorithm with the selected start and goal nodes pass if path is None: # display a error message indicating that no path was found pass else: # Display result st.subheader("Search Result") st.write( f"Algorithm: {algorithm}" ) st.write( f"Solution Path: {' → '.join(path)}" ) st.write( f"Total Path Cost: {cost:.2f}" ) # Visualize NetworkX graph G = nx.DiGraph() for node, neighbors in hospital_graph.items(): for neighbor, weight in neighbors.items(): pass pos = locations fig, ax = plt.subplots( figsize=(10, 6) ) # WRITE REMAINING NETWORKX VISUALIZATION CODE HERE ax.set_title( f"{algorithm} Solution Path" ) ax.axis("off") st.pyplot(fig)
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